

SHIELD-AI

A Formally-Verified Online Learning Controller for Safe Kubernetes Auto-Scaling and Self-Healing

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The locked thesis sentence

"Naive ML-based Kubernetes controllers are unsafe under burst load. SHIELD-AI combines online ML (River) with a formally-verified safety shield (TLA+) to retain ML adaptability while provably satisfying safety invariants that bare controllers violate."

Source: [tasks/THESIS.md:3-7](#)

Why now? The Day-15 motivating failure

	Scaling lag (s)	Error rate (%)	Replicas (start -> end)
	15	0.0	2 -> 6
	5	0.0	2 -> 2
eld)	90	69.2	2 -> 2

The ML-only operator was WORSE than HPA under burst load: it emitted a single heal action and never scaled.

Source: docs/thesis/02_introduction.md:30-37; docs/thesis/07_results.md

Two defects caused the Day-15 failure

- * Ordering bug: decision engine checked heal before scale. Under burst load, anomaly detector flagged high p95 as anomalous, engine emitted heal, scale was never called.
- * No online learning: `_run_online` never called `.learn_one()`. The Day-7 offline model was frozen for 8 days, never adapting to live traffic.
- * Combined effect: ML-only controller stuck at 2 replicas with 69% error rate while HPA scaled correctly.

Source: `src/decision/decision_engine.py:50-95` (P1 fix)

SHIELD-AI architecture

- * Prometheus (10 s) -> Kafka (k8s-metrics) -> Faust 30-s windows
- * -> Kafka (k8s-features) -> Decision Engine (River HTR + HalfSpaceTrees)
- * -> Safety Shield (TLA+ verified, 6 invariants) -> Kafka (k8s-decisions)
- * -> Operator (Kafka consumer + K8s actuator, re-validates)
- * -> workload-v2 Deployment (scale or heal)

Source: docs/thesis/05_proposed_system.md:9-51; docs/paper/main.tex

Why River Hoeffding Adaptive Tree Regressor?

- * Online learning on a stream: one-pass, $O(\text{memory})$, no GPU, no minibatch
- * Concept drift: tree adapts split criteria online; a frozen neural net does not
- * Explainability: leave-one-out perturbation gives defensible answer to "why did the controller scale here?"
- * Rejected alternatives:
 - * - scikit-learn Random Forest (offline; would need full retrain on drift)
 - * - PyTorch neural net (GPU unavailable on single-node kind cluster)
 - * - RL (PPO on replica count; out of M.Tech scope, needs simulation env)

Source: docs/thesis/05_proposed_system.md:115-130

Decision rule - load-first ordering (P1 fix)

```
if predicted_replicas != current_replicas:  
    action = 'scale' # load is dominant  
elif anomaly_score > 2 * threshold:  
    action = 'heal'  
else:  
    action = 'noop'  
  
# after every decision:  
engine.learn(features, current_replicas) # online learn loop
```

Source: src/decision/decision_engine.py:50-95

Safety Shield - the trust boundary

Invariant	What it enforces
SafetyMinReplicas	replicas ≥ 1
SafetyMaxReplicas	replicas ≤ 10
SafetyScalingStep	$ \text{delta} \leq 2$ per decision
SafetyHealNoScale	heal preserves replicas
SafetyBoundedRate	cooldown elapsed or no action pending
LivenessEventuallyScaleUp	sustained demand $\rightarrow$ scale eventually

Source: *specs/SafetyShield.tla:230-290; src/safety/safety_shield.py*

ML+Shield composition - the central paper claim

"The closed-loop system is safe iff the shield is safe, regardless of ML oracle behavior."

- * ML oracle modeled as thin non-deterministic abstraction: any (action, target) in $0..ML_OUTPUT_RANGE$, including out-of-bounds and over-large-step.
- * Two parallel paths in one module: SHIELD (production) and ML_Only (the bug).
- * TLC verifies all six shield invariants hold on every reachable state of the joint spec AND the ML_Only path CAN violate MISafetyMaxReplicas (3-step counterexample).
- * Result: shield is NECESSARY and SUFFICIENT for safety.

Source: [specs/ML_Composition.tla](#); [tlc_run_ml_composition.txt](#); [tlc_run_ml_only_counterexample.txt](#)

Shield in action - offline replay (225 decisions)

Outcome	Count	%
Allowed unchanged	178	79.1%
Clamped ($ \delta > 2$)	47	20.9%
Rejected	0	0.0%
Total	225	100%

- * All 47 clamps are 5-10 -> 4 via max_scale_step=2.
- * Shield never had to reject: ML already conservative (predicts 7 not 10).
- * But clamping is the safety guarantee - no over-large step reaches K8s.

Source: `scripts/replay_shield.py`; `data/features_v2.csv`

Live demo: pipeline + safety shield active

Event	Detail
Producer	publishes workload-v2 metrics every 10s to k8s-metrics
Faust	consumes k8s-metrics, emits 30-s windowed avg to k8s-features
Decision engine	consumes k8s-features, emits scale to k8s-decisions
Actuator	consumes k8s-decisions, validates, applies to Deployment
Load (Locust)	120 users, 941 req/s burst, 100% fail (DB contention)
Result	workload-v2 scaled 2 -> 1 -> 3 -> 5 over ~3 min
Shield rejects	every other action due to cooldown (30s remaining)

Source: logs/operator_actions_demo.log; logs/decisions_demo.log

Why TLA+, not just unit tests?

- * Unit tests: verify that the code passes test cases.
- * TLA+: verify that the code is correct for every possible input (within bounds).
- * Difference: 'tested' vs 'verified' - the shield's invariants hold on every reachable state, not just on the sample the tests happen to cover.
- * Composition spec extends this: shield's safety holds for every possible ML oracle output, not just the ones in our test set.
- * Industrial precedent: Amazon (DynamoDB, S3), Microsoft (Cosmos DB).

Source: docs/thesis/03_literature_survey.md:30-50

Why Kafka, not in-memory bus?

- * Durability: if decision engine crashes, no metrics are lost.
- * Audit trail: every decision is on a Kafka topic (logs/decisions.log).
- * Replay: offline replay (scripts/replay_shield.py) reads from the same topic.
- * Cost: ~10 ms per hop latency. Trade-off accepted for the safety claim.
- * Alternative rejected: in-memory queue - no audit, no replay, no recovery.

Source: docs/thesis/08_discussion.md:8-15

Why Kafka actuator, not Kopf CRD handler?

- * Simpler test surface: one process to test, not two (Kopf + CRD lifecycle).
- * No requeue logic, no watcher, no CRD reconciliation.
- * Decouples the operator from the cluster's CRD registry.
- * Easier to formally model: operator's input is a Kafka topic.
- * Decision documented in AMENDMENTS 2026-08-23.

Source: `src/kopf_operator/actuator.py`; `tasks/AMENDMENTS.md`

Reproducibility - the single-command demo

```
# On a fresh VM:
git clone git@github.com:personal:sudo-Harshk/k8-auto-scaling-self-healing.git
cd k8-auto-scaling-self-healing
make build-image && make load-image
make deploy-kafka deploy-prometheus deploy-workload
# port-forward Prometheus + Kafka (compose uses network_mode: host):
kubectl port-forward svc/kube-prometheus-stack-prometheus 9090:9090 -n monitoring &
kubectl port-forward svc/kafka 9094:9094 -n kafka &
kubectl port-forward svc/workload-v2 8080:8080 -n workload-v2 &
make pipeline-up
docker run ... locust -f scripts/locustfile_v2.py --host=http://localhost:8080
```

Source: Makefile; docs/GOLDEN_RUN.md; ops/compose/pipeline.yaml

Stats & tests (N=10 final evaluation)

- * 53 unit tests pass on every commit (under 1 second).
- * 17 anti-drift tests on the Safety Shield - each intentionally violates one invariant and verifies the class catches it.
- * N=10 head-to-head evaluation: 120 trials (10 seeds x 4 operators x 3 scenarios).
- * Effect sizes with Cohen's d and 95% CI on every comparison metric.
- * TLC: SafetyShield 273,702 states, ML_Composition 7.3M states, all clean.
- * ML_Only counterexample: produces 3-step violation trace (proves shield needed).

Source: tests/; results_N10/; specs/tlc_run_.txt*

Threats to validity

- * Single-node kind cluster (no multi-node scheduling realism).
- * Single workload class (workload-v2, DB-backed Flask + SQLite).
- * 60-second cooldown limits scaling agility.
- * Detection rate 55-65% on organic baseline (anomaly detector tuned for fault-injection patterns).
- * SHIELD-AI cold-start under-predicts on early ramp-up rows.
- * Single Kafka as point of failure.

Source: docs/thesis/07_results.md (Threats); docs/paper/main.tex

Where we sit vs HPA, KEDA, FIRM

	KEDA	FIRM	SHIE
	+	+	+
	X	X	+
	X	X	+ (6 i
	X	X	+ Riv
	X	X	+

Source: docs/thesis/03_literature_survey.md:55-60

Three contributions (recap)

- * Hybrid ML + formal safety controller - K8s operator whose action space is the intersection of ML-driven decisions and a TLA+-verified invariant set.
- * Empirically-validated failure mode of pure ML controllers - Day-15 N=3 evidence showing AI without the shield gets stuck at 2 replicas with 69% error (the motivating failure).
- * Reproducible artifact - 53 unit tests, containerized make demo, full TLA+ TLC trace, FIRM-style ML baseline, live pipeline with Kafka actuator.

Source: tasks/THESIS.md:9-13

Future work & questions

- * Production deployment: multi-node cluster, multi-tenant policies, HA Kafka.
- * Concept drift monitoring: ADWIN on residual error of the replica predictor.
- * Tighter TLA+: probabilistic safety bounds, multi-tenant fairness, leader-election fault tolerance.
- * Beyond K8s: GPU allocation, edge workload placement, network QoS - same architecture (Kafka + Faust + River + TLA+ shield).
- *
- * Questions?

Source: docs/thesis/09_conclusion.md:51-65